

Pramo Jewels

Courier API Integration

Document ID: PJ-CAI-001

Version: 1.0

Status: Approved for Development

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define technical integration standards for courier service APIs.
Audience	Backend Developers, DevOps, QA, Logistics Team
Version	1.0

Courier API Capabilities

API Capability	Description
Authentication	Secure API keys/OAuth stored in secret manager
Serviceability	PIN code availability lookup
Rate Calculation	Fetch shipping charges and delivery estimates
Shipment API	Create shipments and courier orders
Pickup API	Schedule, reschedule or cancel pickups
Tracking API	Retrieve real-time shipment status
Webhook API	Receive shipment events securely
Label API	Generate AWB and printable shipping labels

Integration Architecture

Use a provider adapter layer to abstract courier-specific APIs and expose a unified internal shipping interface.

API Workflow

Authenticate → Validate Serviceability → Create Shipment → Generate AWB → Schedule Pickup → Track Shipment → Receive Webhooks → Update Order Status.

Error Handling

Retry transient failures with exponential backoff, log permanent failures, and queue failed requests for manual review when required.

Security

Enforce HTTPS/TLS, verify webhook signatures, rotate API credentials regularly, apply IP allowlists where supported, and protect secrets using centralized secret management.

Performance

Use asynchronous processing for shipment creation and tracking updates, cache serviceability lookups, and respect courier API rate limits.

Monitoring

Monitor API latency, success rate, webhook delivery, rate-limit responses, failed requests, and courier availability.

Integrations

Integrates with Order Management, Inventory, Notifications, Customer Accounts, Analytics, Audit Logging and Shipping Partner modules.

Acceptance Criteria

Courier APIs integrate reliably, shipment data remains synchronized, failures are recoverable, and all API interactions are traceable.